

Data Science

Lecture 01: Introduction

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Module 1

The Data Deluge

Guiding Question: *We generate more data than we can store. Does having more of it make us know more?*

How Much Data Exists?

Year	Global datasphere	Per person on earth
2002	5 exabytes of new information	—
2010	~1 zettabyte	a trillion gigabytes in total
2020	~40 zettabytes	≈ 5,200 GB
2025	~175 zettabytes	≈ 21,000 GB

In 2007 the amount of information produced first **surpassed available storage capacity**. Since then, most data that is generated is never stored.

Growth Is Not the Interesting Part

Volume

More bytes than anyone can read, index or curate.

Heterogeneity

Text, sensors, images, logs, transactions — rarely in a common schema.

Velocity

Streams that must be acted on before they can be stored.

The scarce resource is not data. It is **the ability to turn data into justified conclusions** — which is what this course is about.

Science at Scale

Large Hadron Collider

1 MB per event, 600 million events per second — roughly **18 zettabytes per year** if everything were kept.

A single tier-1 centre stores ~2.2 petabytes per year.

The filter is the experiment

Almost all events are **discarded in hardware**, in real time, by trigger rules written in advance.

What is kept is decided by a theory of what matters.

Data at Web Scale

Search

Web-scale indexing processes petabytes per day and answers in milliseconds.

Platforms

Interaction logs — clicks, dwell time, scrolls — dwarf the content itself.

Models

Training corpora for language models are measured in trillions of tokens.

In every case the value comes from **aggregation across many weak signals**, not from any single record.

Data is never raw: at the LHC a physics theory decides what is recorded, and on the web value comes from aggregating signals that are individually worthless. **Someone always chose what the dataset contains.**

Module 2

Does Data Speak for Itself?

Guiding Question: *Twenty-nine teams analysed the same dataset and the same question. Why did they not get the same answer?*

"The End of Theory"

With enough data, the numbers speak for themselves. Correlation supersedes causation, and science can advance without coherent models or unified theories.

Chris Anderson, *Wired*, 2008. The argument has returned, almost verbatim, in claims that sufficiently large models make domain theory unnecessary.

Is it right?

One Dataset, Many Answers

Twenty-nine research teams received **the same dataset** and the same question:

Are football referees more likely to give red cards to dark-skinned players than to light-skinned players?

The result

Odds ratios ranged from **0.89 to 2.93**.
Twenty teams found a significant effect
nine did not.

The reason

No team made an error. They made
different defensible modelling choices
— which covariates, which exclusions,
which model family.

The data did not determine the answer. The **analysis** did. (Silberzahn et al. 2018)

Correlation Is Not Causation

Spurious correlations are easy to find and get **easier as datasets grow**: with enough variables, some pair will track another closely by chance alone.

Why it happens

Testing many pairs makes an extreme correlation likely *somewhere*, even when every pair is independent.

Why more data does not fix it

More rows shrink the confidence interval around a **meaningless** number. The estimate gets more precise, not more valid.

Summary Statistics Hide the Data

Datasets whose **summary numbers agree exactly** — same averages, same spread, same strength of association — can look completely different when plotted: a dinosaur, a star, a set of lines (Matejka & Fitzmaurice 2017).

We will make "average", "spread" and "association" precise in Lecture 02, and meet this phenomenon again there as Anscombe's quartet.

Plot the data. Then plot it again, differently.

Where This Leaves Us

Data does not speak for itself

Every dataset was selected, filtered and encoded by someone.

Data cannot be read intuitively

Human intuition fails at scale, in high dimensions, and with base rates.

So theory is needed *more*

Models make the assumptions explicit enough to be criticised.

Big data does not end theory. It **raises the cost of implicit theory**, because assumptions that stay unstated now get automated and scaled.

Module 3

What Is Data Science?

Guiding Question: *If data science is not statistics, not machine learning and not engineering, what is left that is its own?*

A Definition

Data Science researches and applies technologies, systems and algorithms to gain knowledge from potentially large, heterogeneous and dynamically changing data sets — for a particular domain. (Dhar 2013)

The phrase doing the work is "**for a particular domain**": without it, there is no criterion for which patterns count as knowledge.

Neighbouring Fields

Machine Learning

Learning from past experience to improve at a task in the future.

Artificial Intelligence

Building systems that behave intelligently.

Data Engineering

The twin field: managing, organising and providing access to data.

Data science is not a subset of any of them. It is the discipline of getting from a **domain question** to a **defensible answer**, using whichever of these tools fits.

Three Kinds of Task

Explorative

Find patterns.

Which customers behave similarly?

Descriptive

Summarise what is there.

What does an average customer spend?

Predictive

Anticipate future outcomes.

Is this transaction fraudulent?

Naming the task first constrains everything else — the method, the evaluation measure, and what counts as success.

Data science is defined by its **question and its domain**, not by its tools. Explorative, descriptive and predictive tasks demand different methods and different evidence — so the first decision in any project is which one you are doing.

Module 4

How Knowledge Is Acquired

Guiding Question: A pattern in data is not yet knowledge. What has to happen in between?

Three Modes of Reasoning

	From	To	What it guarantees
Deduction <i>(logic)</i>	premises	what must follow	certain — but adds no new content
Induction <i>(statistics)</i>	observed patterns	a regularity , a hypothesis	nothing — content-adding, not guaranteed
Abduction <i>(probability)</i>	observations	the best available explanation	nothing — only comparable against rivals

Machine learning lives in the bottom two rows.



The Scientific Reasoning Cycle

The three modes are not alternatives — they take turns.

Example: gravity.

Deduction

From the theory: all objects of equal mass fall at the same rate everywhere on earth.

Induction

Design an experiment, measure fall times at several locations, compute means and standard deviations.

Abduction

Times agree at one location but vary between locations.
Best explanation: distance from earth's mass centre.

The cycle turns: the abduced explanation becomes the next theory to deduce from. A data-science project runs the same loop, usually several times.

Mini-Exercise

A dashboard shows that customers who use the mobile app churn **less** than those who do not.

- 1 What would a purely **inductive** reading conclude?
- 2 Give two different **abductive** explanations.
- 3 What **deduction** would let you distinguish them?

Mini-Exercise

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INDUCTION

"App use reduces churn" — a regularity read straight off the data.

ABDUCTION

The app causes loyalty — *or* loyal customers install the app. Same correlation, opposite causal direction.

DEDUCTION

If the app causes loyalty, randomly prompting a subgroup to install it should lower their churn — an experiment deduced from the hypothesis.

What Counts as Knowledge?

Knowledge discovery in databases (KDD) is the non-trivial process of identifying patterns in data that are

- **valid** — they hold beyond the sample,
- **novel** — they were not already known,
- **relevant** — they matter for a decision,
- **interpretable** — they can be understood and acted upon.

A pattern that fails **any** of the four is not knowledge. Most patterns a large dataset offers fail at least one.

≡ Knowledge is a pattern that is **valid, novel, relevant and interpretable**. Data mining produces patterns only the four criteria turn one into knowledge — and most patterns fail at

least one of them.

Module 5

The Process and the Craft

Guiding Question: *Most of a data-science project is spent before any model is fitted. On what, and why does it matter?*

Step 1 — Selection of Data

Types of source

- **Unstructured:** flat files, documents, text and multimedia
- **Structured:** relational databases, XML, web APIs

Types of data

media · object-relational · spatial and spatio-temporal · graphs and networks · time series and sequences · data streams

Selection is where the **first and largest** bias enters. What was not selected cannot be discovered later.

Step 2 — Preprocessing

Selected

manually or
automatically, e.g. by
crawling

Cleaned

errors and noise
removed — *garbage
in, garbage out*

Integrated

combined with other
sources, resolving
conflicts

Transformed

into a reusable
representation for
analysis

This step routinely consumes most of a project's effort — and every choice in it is a modelling decision, not a clerical one.

Steps 3–5 — Mining, Evaluation, Deployment

Data mining

Apply the algorithm — clustering, classification, rule mining — that fits the task.

Evaluation

Test the patterns against the four criteria: valid, novel, relevant, interpretable.

Deployment

Put the result where a decision is made, and monitor it there.

The process is a **loop**, not a pipeline. Evaluation routinely sends you back to selection.

What the Work Actually Looks Like

Ask

Turn a domain question into something a dataset could answer — and check that it is the question worth answering.

Build

Acquire, clean, model, evaluate. Iterate, mostly backwards.

Communicate

Deliver a result someone can act on, with its uncertainty attached.

Three Skill Areas

Domain knowledge

What the data means, which values are impossible, which differences matter.

Statistics and modelling

What may be inferred, with what uncertainty, under which assumptions.

Engineering

Getting the data, at scale, reproducibly, into production.

The interesting failures happen **between** the three: a technically flawless model of the wrong quantity, or a correct analysis nobody can deploy.

The process is a **loop, not a pipeline**, and it spans three skill areas that nobody holds equally. The characteristic failures happen at the seams — a flawless model of the wrong quantity, or a correct analysis nobody can deploy.

Module 6

Applications, and Their Failures

Guiding Question: What happens to a model's validity once the world can see the model?

Web and Social Media

Google Flu Trends

Search queries were used to nowcast influenza faster than public health reporting.

And then it failed

It overestimated cases substantially, because search behaviour changed — partly *because of media coverage of the tool itself*.

A model that acts on the world **changes the world it measures**. Validity is not a property a model keeps once earned.

Text and Language

Classic tasks

Summarisation, entity extraction, sentiment analysis, question answering — each once a research programme of its own.

Today

A single pretrained model handles all of them, with the task specified in words rather than in code.

What did **not** change: the need to evaluate on data the model has not seen, and to state what the output may be used for.

Business and the Public Sector

Reported potentials from the era of the first big-data wave: large retail revenue gains, hundreds of billions in healthcare savings, comparable figures for e-government.

These are **projections from consultancy reports**, not measured outcomes. Treat them as claims to be interrogated — whose baseline, whose costs, and what happened afterwards?

Applications show what is possible their failures show what must be checked. Ask of every deployed model: what changed in the world because it exists?

Image Credits & References

Dhar, Vasant (2013). Data science and prediction. *Communications of the ACM*.

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